Delta Sleep Inducing Peptide (DSIP) Available for Research Use Only

100 µg–0.5 mg SC or IN Near Bedtime

Delta sleep-inducing peptide (DSIP) is a synthetic nine-amino-acid peptide based on a sequence originally isolated from rabbit cerebral venous blood during experiments associated with slow-wave sleep. Small human intravenous studies of DSIP produced mixed findings, ranging from improved sleep measures to weak or clinically limited effects. Community reports are mixed: some users describe improved sleep, while others report no effect. Typical community subcutaneous doses are substantially smaller than the intravenous trial dose and probably produce lower systemic exposure.

💉 💉 **Short-term safety is likely. Long-term safety and repeated subcutaneous use are speculative.**

🅾️ **Sleep efficacy is inconclusive. Human studies report mixed findings, ranging from improved sleep measures to effects considered weak and of limited therapeutic significance.**

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Deep sleep, also called slow-wave sleep or stage N3, is the deepest non-rapid eye movement (non-REM) sleep stage. It is characterized by high-amplitude delta activity, a high arousal threshold, and association with restorative processes and pulsatile growth hormone release. Rapid eye movement (REM) sleep features rapid eye movements, vivid dreaming, skeletal muscle atonia, and cortical activity involved in emotional processing and memory. Lighter non-REM stages N1 and N2 are more easily interrupted and contribute to normal sleep cycling and memory processing.

Small human studies of DSIP support short-term tolerability of slowly administered intravenous DSIP. Long-term safety and the safety of repeated subcutaneous use are likdely. Widespread community use provides only a reassuring indirect safety signal.

Human evidence for sleep improvement is mixed. Several small early intravenous studies reported improvements in sleep duration, efficiency, or continuity. A 1987 placebo-controlled, double-blind trial using intravenous DSIP for seven successive nights reported improvements in sleep and daytime function. A 1992 double-blind trial using three consecutive intravenous doses

found only weak objective improvements and no subjective sleep-quality benefit. Overall efficacy remains inconclusive, and community reports are also mixed.

Dosing Evidence and Patterns

DSIP is rapidly degraded in blood, which limits its persistence after administration. Human sleep studies used slow intravenous (IV) administration and reported that **the slow rate was important to the observed response**. Subcutaneous (SC) administration is now common in the research-use community. It could plausibly slow absorption relative to intravenous administration.

Schneider-Helmert et al. (1981) and Schneider-Helmert and Schoenenberger (1983) describe human experiments using slowly administered intravenous DSIP at 25 nmol/kg. Based on DSIP's molecular weight of approximately 848.8 g/mol, this equals about 21.2 µg/kg, or approximately 1.5 mg for a 70 kg person.

Graf and Kastin (1984) summarized a U-shaped relationship between DSIP's effects and both dose and administration timing across the early experimental literature.

Human subcutaneous bioavailability and pharmacokinetics have not been established for DSIP. In the absence of DSIP-specific evidence, a broad 30%–80% bioavailability range observed across other subcutaneously administered peptides can be used as an illustrative modeling assumption. Within this range, matching the total systemic exposure from 1.5 mg administered intravenously would require approximately 1.9–5 mg administered subcutaneously. This calculation is speculative and does not establish dose equivalence because DSIP absorption, degradation, peak concentration, and tissue exposure may differ from those of other peptides.

Published human studies examined several short intravenous regimens, including three, four, seven, and ten consecutive night series. Monthly, intermittent, or as-needed repetition schedules used by contemporary community researchers remain speculative community practices.

Common Community Dosing

Community discussions commonly describe 100 µg–0.5 mg near bedtime, with 250 µg frequently cited. Both subcutaneous and intranasal administration are reported. These doses and routes are speculative and are not derived from human trials.

Interestingly, a **subcutaneous doses from 1.5 mg-5 mg which would more closely resemble the IV doses used in human trials are almost never discussed in peptide communities**.

Some users report continuous use, while other users reported cycles like four to eight weeks ON followed by four to eight weeks OFF, sometimes combined with five days ON and two days OFF each week. All schedules are speculative community practices without evidence.

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Human intravenous studies reported sleep-related effects beginning within approximately one hour and, in some experiments, delayed effects on subsequent sleep. Contemporary **users commonly administer subcutaneous DSIP shortly before bedtime.** A slower absorption profile after subcutaneous administration is plausible, while any specific pre-sleep interval, onset time, or duration remains speculative.

Benefits

Reported effects vary substantially across studies and individuals. Small human trials range from measurable improvements in sleep to weak or no effects. Differences in study design, population, administration route, dose, timing, and individual response could all contribute. Community reports are similarly mixed.

- **Sleep architecture:** Effects are inconclusive. Some human studies reported improved sleep efficiency, shorter sleep latency, or better sleep continuity, while other controlled findings were weak or nonexistent. Effects on slow-wave sleep vary across human and animal studies.
- **Neuroendocrine modulation:** Intravenous DSIP reduced plasma adrenocorticotropic hormone-like immunoreactivity (ACTH-LI) for at least three hours in one small randomized crossover study, without changing cortisol. Broader modulation of hypothalamic-pituitary-adrenal (HPA) axis function and stress-related arousal is plausible.
- **Daytime function:** Small human studies reported improved daytime alertness and performance after intravenous DSIP, with mood improvement reported in one small open study.
- **Subjective recovery:** Improved physical recovery is reported by some community users and remains speculative.

Potential Applications

- **Insomnia or fragmented sleep:** Experimental intravenous use has mixed human evidence, with benefits reported in some small studies and weak effects in another controlled trial.
- **Stress-related sleep disruption:** Benefits are plausible based on limited human neuroendocrine findings and preclinical evidence.
- **Recovery support:** Use to support recovery through improved sleep remains speculative and is primarily community reported.
- **Non-benzodiazepine sleep research:** DSIP is an investigational non-benzodiazepine approach to sleep modulation; comparative advantages remain speculative.

Cautions

- **Pregnancy and breastfeeding:** Safety is unknown, so use warrants particular caution.

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- **Severe or unstable psychiatric illness:** Effects on sleep, arousal, and psychiatric symptoms are uncertain.
- **Concurrent sedative-hypnotics, central nervous system depressants, or other investigational sleep agents:** Additive effects on sedation, arousal, or sleep architecture are plausible.
- **Severe hepatic or renal impairment:** Effects on DSIP exposure and clearance are unknown.

Reported and Possible Side Effects

- **Short-term tolerability:** Small human intravenous studies generally described DSIP as well tolerated during brief exposure.
- **Headache, nausea, or vertigo:** These effects were reported occasionally in the older clinical literature.
- **Vivid dreams or morning grogginess:** These are community-reported effects and remain speculative.
- **Autonomic changes during anesthesia:** In one small randomized study, intravenous DSIP administered with isoflurane increased heart rate and reduced heart-rate variability.

Diminished Response and Cycling

Waning responsiveness during repeated subcutaneous use remains speculative. DSIP has no identified specific receptor, and human data on immunogenicity or downstream physiological adaptation are unavailable. On-off cycling is a speculative community practice intended to preserve responsiveness.

Use with Low Dose Epitalon

Low-dose Epitalon for sleep is a community practice. Practitioners commonly discuss 100–300 µg, often administered intranasally before bed, with 100 µg frequently cited. This dose, route, and schedule are speculative.

Epitalon has a plausible mechanistic connection to circadian regulation based on preclinical evidence involving pineal signaling and evening melatonin. Sleep benefits from low-dose nightly administration remain speculative.

Combining low-dose DSIP with low-dose Epitalon is also a speculative community practice. Any additive benefit, dosing pattern, or cycling schedule remains speculative.

Low dose Pinealon is another speculative addition to DSIP, often combined with Epitalon.

Biological Mechanism

DSIP-like immunoreactivity has been detected in mammalian tissues and fluids, but the identity, synthesis, and physiological role of an endogenous DSIP molecule remain unresolved. DSIP has

no established molecular mechanism and no identified specific receptor. Experimental studies report effects on sleep-related electrophysiology, neuroendocrine signaling, cellular stress responses, and autonomic activity. These findings support several plausible, context-dependent mechanisms rather than a single validated pathway.

Neural signaling: In rat neurons, DSIP potentiated GABA-activated currents and modified N-methyl-D-aspartate (NMDA)/glutamate signaling. Animal electroencephalogram (EEG) studies also reported increased delta-frequency activity in some models. Modulation of inhibitory-excitatory balance is therefore a plausible mechanism for DSIP's sleep-related effects in humans.

Stress-axis signaling: Preclinical studies and one small randomized human crossover study indicate that DSIP can influence adrenocorticotropic hormone-related signaling. In the human study, ACTH-like immunoreactivity decreased while cortisol remained unchanged, suggesting an effect upstream of adrenal cortisol output. Modulation of hypothalamic-pituitary-adrenal signaling is a plausible mechanism for stress-related effects.

Cellular stress and neuroprotection: In rat models, DSIP reduced neurological impairment and mortality after cerebral ischemia and preserved brain-mitochondrial oxidative phosphorylation during hypoxia. Other rat studies found modulation of glutamate excitation and lipid-peroxidation markers. Neuroprotective activity in ischemic or hypoxic settings is likely, while broader recovery and resilience benefits are plausible.

Autonomic signaling: In a small randomized anesthesia study, intravenous DSIP increased heart rate and reduced heart-rate variability during isoflurane anesthesia, consistent with reduced parasympathetic tone. Effects on autonomic regulation during ordinary sleep are plausible.

References

1. Schoenenberger GA, Monnier M. Characterization of a delta-electroencephalogram (-sleep)-inducing peptide. Proc Natl Acad Sci U S A. 1977;74(3):1282–1286. doi:10.1073/pnas.74.3.1282.

2. Schoenenberger GA, Maier PF, Tobler HJ, Wilson K, Monnier M. The delta EEG (sleep)-inducing peptide (DSIP). XI. Amino-acid analysis, sequence, synthesis and activity of the nonapeptide. Pflugers Arch. 1978;376(2):119–129. doi:10.1007/BF00581575.

3. Schneider-Helmert D, Graf M, Schoenenberger GA. Synthetic delta-sleep-inducing peptide improves sleep in insomniacs. Lancet. 1981;1(8232):1256–1257. doi:10.1016/S0140-6736(81)92417-X.

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4. Schneider-Helmert D, Gnirss F, Monnier M, Schenker J, Schoenenberger GA. Acute and delayed effects of DSIP (delta sleep-inducing peptide) on human sleep behavior. Int J Clin Pharmacol Ther Toxicol. 1981;19(8):341–345. PMID:6895513.

5. Schneider-Helmert D, Schoenenberger GA. The influence of synthetic DSIP (delta-sleep-inducing-peptide) on disturbed human sleep. Experientia. 1981;37(9):913–917. doi:10.1007/BF01971753.

6. Schneider-Helmert D, Schoenenberger GA. Effects of DSIP in man. Multifunctional psychophysiological properties besides induction of natural sleep. Neuropsychobiology. 1983;9(4):197–206. doi:10.1159/000117964.

7. Kaeser HE. A clinical trial with DSIP. Eur Neurol. 1984;23(5):386–388. doi:10.1159/000115717.

8. Graf MV, Kastin AJ. Delta-sleep-inducing peptide (DSIP): a review. Neurosci Biobehav Rev. 1984;8(1):83–93. doi:10.1016/0149-7634(84)90022-8.

9. Schoenenberger GA. Characterization, properties and multivariate functions of delta-sleep-inducing peptide (DSIP). Eur Neurol. 1984;23(5):321–345. doi:10.1159/000115711.

10. Schneider-Helmert D. Effects of delta-sleep-inducing peptide on 24-hour sleep-wake behaviour in severe chronic insomnia. Eur Neurol. 1987;27(2):120–129. doi:10.1159/000116143.

11. Graf MV, Saegesser B, Schoenenberger GA. Degradation and aggregation of delta sleep-inducing peptide (DSIP) and two analogs in plasma and serum. Peptides. 1987;8(4):599–603. doi:10.1016/0196-9781(87)90031-3.

12. Bjartell A, Ekman R, Bergquist S, Widerlöv E. Reduction of immunoreactive ACTH in plasma following intravenous injection of delta sleep-inducing peptide in man. Psychoneuroendocrinology. 1989;14(5):347–355. doi:10.1016/0306-4530(89)90004-8.

13. Bes F, Hofman W, Schuur J, Van Boxtel C. Effects of delta sleep-inducing peptide on sleep of chronic insomniac patients. A double-blind study. Neuropsychobiology. 1992;26(4):193–197. doi:10.1159/000118919.

14. Shandra AA, Godlevskii LS, Brusentsov AI, et al. Effects of delta-sleep-inducing peptide in cerebral ischemia in rats. Neurosci Behav Physiol. 1998;28(4):443–446. doi:10.1007/BF02464804.

15. Lysenko AV, Alperovich DV, Uskova NI, Mendzheritsky AM. Metabolic features of the adaptive effect of delta-sleep inducing peptide and piracetam under hyperoxic conditions. Biochemistry (Mosc). 1999;64(6):652–657. PMID:10395980.

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16. Stanojlović OP, Zivanović DP, Susić VT. The effect of delta sleep-inducing peptide on the EEG and power spectra in rat. Indian J Physiol Pharmacol. 2000;44(4):428–434. PMID:11214497.

17. Sibarov DA, Kovalenko RI, Malinin VV, Khavinson VKh. Epitalon influences pineal secretion in stress-exposed rats in the daytime. Neuro Endocrinol Lett. 2002;23(5-6):452–454. PMID:12500171.

18. Khvatova EM, Samartzev VN, Zagoskin PP, Prudchenko IA, Mikhaleva II. Delta sleep inducing peptide (DSIP): effect on respiration activity in rat brain mitochondria and stress protective potency under experimental hypoxia. Peptides. 2003;24(2):307–311. doi:10.1016/S0196-9781(03)00040-8.

19. Kovalzon VM, Strekalova TV. Delta sleep-inducing peptide (DSIP): a still unresolved riddle. J Neurochem. 2006;97(2):303–309. doi:10.1111/j.1471-4159.2006.03693.x.

20. Grigor'ev VV, Ivanova TA, Kustova EA, Petrova LN, Serkova TP, Bachurin SO. Effects of delta sleep-inducing peptide on pre- and postsynaptic glutamate and postsynaptic GABA receptors in neurons of the cortex, hippocampus, and cerebellum in rats. Bull Exp Biol Med. 2006;142(2):186–188. doi:10.1007/s10517-006-0323-9.

21. Pomfrett CJ, Dolling S, Anders NR, Glover DG, Bryan A, Pollard BJ. Delta sleep-inducing peptide alters bispectral index, the electroencephalogram and heart rate variability when used as an adjunct to isoflurane anaesthesia. Eur J Anaesthesiol. 2009;26(2):128–134. doi:10.1097/EJA.0b013e32831c8644.

22. Araj SK, Brzezik J, Mądra-Gackowska K, Szeleszczuk Ł. Overview of Epitalon—Highly Bioactive Pineal Tetrapeptide with Promising Properties. Int J Mol Sci. 2025;26(6):2691. doi:10.3390/ijms26062691.

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